

Motivation: Why Extend OT?

Limitation	Extension	Examples
2 measures only	Multimarginal OT	Barycenters
Equal total mass	Unbalanced OT	Cell data, noisy histograms
Fixed coupling structure	OT + constraints	Finance (Martingale)
Same ambient space	Gromov–Wasserstein	Shape / graph matching
Scalar measures	Vector/matrix OT	Tensors, covariances

All preserve the variational flavor of OT!

OT Atlas

Monge OT $\xrightarrow[\textit{allow mass splitting}]{\textit{Relax formulation:}}$ Kantorovich OT $\xrightarrow[\textit{entropy regularizer}]{\textit{Smooth with}}$ Entropic OT (Sinkhorn)